TenderLens: An Evidence-Linked Compliance Review Agent for Public Tender Responses

ELEC5623 Group Business Proposal · Track A — AI Agent Product

Group 7 — Mia Chen, Liam O'Brien, Priya Raman, Tom Nguyen, Sara Haddad

Cover information	
Track	A — AI Agent Product
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Lab tutor and approval	Dr K. Morrison — approved in the Week 4 lab, 26 August 2026: "Scope limited to the mandatory-compliance stage; approved."
Product vision	Every compliance decision on a tender response is backed by a verified quotation the panel can see in one click.

1. Executive Summary

Procurement officers in Australian public agencies evaluate tender responses against published conditions of participation, mandatory requirements and weighted evaluation criteria. A single request for tender (RFT) commonly produces between five and twenty supplier responses of 40 to 150 pages each, and every response must be checked for mandatory compliance before it can be scored. The Commonwealth Procurement Rules require that submissions which do not meet a condition for participation are excluded and that the reasons are documented (Department of Finance, n.d.). In practice this means an evaluation panel member reads each response searching for evidence of dozens of requirements, records where each requirement is addressed and justifies every "non-compliant" finding.

This proposal presents **TenderLens**, a Track A AI Agent Product that supports the compliance-checking stage of tender evaluation. TenderLens does not score tenders and does not recommend a supplier. It (1) extracts a structured requirement register from the RFT, (2) parses each supplier response while preserving page and section locations, (3) locates candidate evidence for each requirement, (4) asks a language model to classify each requirement as *addressed*, *partially addressed* or *not found* with verbatim quotations, (5) verifies every quotation against the source document and (6) presents an evidence matrix that the panel member confirms or overrides. Every finding links to the exact page and paragraph it relies on.

Generative AI is needed because requirements and responses rarely share vocabulary: an RFT may require "a documented business continuity arrangement", while a supplier describes "our tested disaster recovery runbook with a four-hour recovery objective". Keyword search fails on such paraphrases, and the judgement of whether a passage *satisfies* a requirement is semantic. The design keeps the model inside a narrow, verifiable role: it may only cite text that exists in the response, and code rejects any citation that does not match the source.

Evaluation compares TenderLens with two baselines — manual review and direct whole-document LLM review without verification — on 24 synthetic tender responses built from three public RFT templates, with requirement-level labels produced by two annotators. The primary measures are requirement-level classification accuracy (macro-F1), citation precision, evidence recall, the rate of unsupported findings and reviewer time per response (Section 8).

2. Introduction and Background

2.1 Application Context

Public procurement in Australia follows a staged process: approach to market, receipt of responses, compliance check, detailed evaluation against weighted criteria, value-for-money assessment and recommendation (Department of Finance, n.d.). The compliance check is a gate. A response that fails a mandatory condition is excluded from further evaluation, so an incorrect exclusion can be challenged by the supplier, and an incorrect inclusion wastes panel effort and exposes the agency to complaint. Agencies therefore expect every exclusion decision to be traceable to the response text.

Our partner context is a mid-sized state agency procurement team (described with their permission in anonymised form) that runs roughly 60 open tenders per year. Panels are typically three people, and each panel member performs an independent compliance check before a moderation meeting.

2.2 Evidence that the Problem Exists

In a structured interview with two procurement officers and one evaluation panel chair, all three described the compliance check as the most repetitive part of evaluation. The officers estimated 45–90 minutes per response for a medium tender with 30–40 mandatory requirements. In a small timing exercise, two of our team members checked three publicly available sample responses against a 32-requirement template; median time was 52 minutes per response, and the two members disagreed on 6 of 96 requirement decisions, mostly where the response addressed a requirement indirectly in an appendix.

These observations are consistent with the broader literature on long-document reading: relevant information is often distributed across sections, and readers — human or machine — miss content that appears in unexpected locations (Liu et al., 2024).

2.3 Current Practice, Existing Tools and Technical Motivation

Current practice relies on spreadsheets and PDF search. E-procurement platforms (for example the AusTender portal for publishing approaches to market) manage lodgement and deadlines but do not assess content. Commercial contract-analysis tools extract clauses from contracts but are designed for legal agreements rather than for matching a supplier's narrative to a requirement list. General-purpose chat assistants can summarise a response, but their summaries are not linked to verifiable locations and may state that a requirement is met without textual support, a known failure mode of large language models (Ji et al., 2023).

Retrieval-augmented generation (Lewis et al., 2020) and citation-generating models (Gao et al., 2023) suggest a better pattern: ground each statement in retrieved text and make the grounding checkable. TenderLens applies this pattern to a setting where checkability is a legal and administrative requirement rather than a convenience.

3. Problem Statement and Motivation

3.1 Specific Problem

Given an RFT's list of mandatory requirements and one supplier response, a panel member must determine for each requirement whether the response addresses it, and record where. The task is laborious because evidence is scattered, vocabulary differs between buyer and supplier, and each decision must be justified.

3.2 Who Is Affected

Evaluation panel members carry the reading load. Procurement managers carry the risk of challenge if exclusions are poorly documented. Suppliers are affected when a compliant response is wrongly excluded.

3.3 Why the Problem Matters

Errors at the compliance gate are costly in both directions. A wrongful exclusion can trigger a complaint under the agency's procurement complaints process and delay contract award; a missed non-compliance can lead to awarding a contract to a supplier who cannot meet a mandatory obligation. Time spent on mechanical searching also reduces time available for the qualitative evaluation that panels are actually expert in.

3.4 Why Generative AI Is Appropriate

The core operation — deciding whether a passage satisfies a requirement written in different words — is a semantic judgement over natural language. Rule-based matching and keyword search cannot recognise that "our ISO 27001-certified information security management system" satisfies "the supplier must maintain a recognised information security framework". At the same time, the risks of generative models (fabricated support, overconfidence) are exactly the risks a compliance process cannot accept. The proposal therefore uses GenAI only where semantic judgement is required, and surrounds it with deterministic verification: quotations must exist in the source, every requirement must receive a decision with evidence or an explicit "not found", and the human reviewer decides.

3.5 Project Objective and Success Hypothesis

Objective: build a working prototype that produces a verified evidence matrix for one RFT and one supplier response within the semester.

Hypothesis: compared with direct whole-document LLM review, TenderLens will reduce unsupported findings to below 5% while keeping requirement-level macro-F1 at or above 0.80, and will reduce panel-member time per response by at least 30% compared with manual review.

4. Stakeholders and Requirements Engineering

4.1 Stakeholders and Needs

Stakeholder	Role	Primary need
Panel member	Checks compliance of each response	Find evidence quickly; justify each decision
Panel chair	Moderates disagreements	See where members' decisions differ and why
Procurement manager	Owns process integrity	Auditable record of every exclusion
Supplier	Submits response	Fair, consistent assessment of what they wrote
Agency IT security	Approves tools	No confidential tender data sent to unapproved services

4.2 Primary User Stories

- **US1** As a panel member, I want each mandatory requirement listed with the passages that address it, so that I can confirm compliance without reading the whole response linearly.
- **US2** As a panel member, I want to see when no evidence was found, so that I can check the response myself before excluding it.
- **US3** As a panel chair, I want to compare two members' confirmed matrices, so that moderation focuses on disagreements.

- **US4** As a procurement manager, I want an exportable record of decisions with page references, so that exclusions can be defended.

4.3 Scope and Boundaries

In scope: one RFT and one or more text-based PDF responses at a time; mandatory requirements expressed as discrete statements; English-language documents; evidence matrix review and export.

Out of scope: weighted scoring and ranking of suppliers; pricing evaluation; scanned image-only documents; integration with AusTender or agency records systems; any automated exclusion decision.

4.4 Assumptions

- RFT requirements can be extracted as a numbered list with at most light manual correction.
- Responses are text-based PDFs; scanned pages are flagged rather than processed.
- A cloud model endpoint approved for unclassified data is available for development; production deployment would require an agency-approved endpoint.

4.5 Functional Requirements

Each member owns at least three functional requirements (responsibility areas in Section 10.2).

ID	Requirement	Priority	Owner	Acceptance criterion
FR1	Extract mandatory requirements from an RFT into a numbered register	Must	A1	≥95% of requirements in three test RFTs extracted with correct text and numbering
FR2	Let the user correct, merge or split extracted requirements before a check	Must	A1	Edits saved and used in the next run; original extraction retained
FR3	Version every prompt and record the version with each finding	Must	A1	Every stored finding carries a prompt version present in the repository
FR4	Parse a response PDF into paragraphs with page and section locators	Must	A2	100% of paragraphs carry page and section metadata
FR5	Flag pages without a text layer instead of processing them	Must	A2	Image-only test pages are listed as "needs manual check"
FR6	Export the confirmed matrix with page references	Should	A2	Export opens in Excel and contains every confirmed decision
FR7	Retrieve candidate passages for each requirement	Must	A3	Evidence recall@8 ≥0.85 against annotated evidence
FR8	Allow one additional agent-initiated search per requirement	Should	A3	Search requests logged; never more than two per requirement
FR9	Run the evaluation harness on a labelled set with one command	Must	A3	Harness reproduces M1–M7 from logged runs
FR10	Classify each requirement as addressed, partial or not found	Must	A4	Macro-F1 ≥0.80 against adjudicated labels
FR11	Attach verbatim quotations to every addressed or partial finding	Must	A4	100% of displayed quotations match the source text
FR12	Reject findings whose quotations do not match the source	Must	A4	Zero unverified quotations reach the reviewer view
FR13	Show each quotation in context with one click	Must	A5	Reviewer reaches the source paragraph in ≤2 interactions
FR14	Let the reviewer confirm or override each finding with a note	Must	A5	Override stored with reviewer, time and original AI finding

ID	Requirement	Priority	Owner	Acceptance criterion
FR15	Compare two reviewers' confirmed matrices	Could	A5	Disagreements listed per requirement

4.6 Non-Functional Requirements

ID	Requirement	Measure and target
NFR1	Grounding	0% of displayed quotations absent from source (verified by code)
NFR2	Latency	≤4 minutes to produce a matrix for a 100-page response with 40 requirements
NFR3	Reproducibility	Prompt version, model identifier and retrieval settings logged for every run
NFR4	Privacy	No response text stored outside the local workspace; provider data retention disabled
NFR5	Usability	System Usability Scale (Brooke, 1996) ≥70 in the reviewer study
NFR6	Cost	≤A$1.50 model cost per response at list prices

4.7 Constraints

ID	Constraint
C1	Only public RFT templates and synthetic responses are used; no real supplier data
C2	The prototype runs locally on a team laptop; the only external call is to the model endpoint
C3	Model spend for the semester is capped at A$150
C4	Eight development weeks, five students

4.8 Requirements-to-Evaluation Traceability

Requirement	Evaluation metric (Section 8.4)
FR1–FR2	M1 requirement extraction accuracy
FR4–FR5	M2 locator completeness
FR7–FR8	M3 evidence recall@8
FR10	M4 macro-F1
FR11–FR12, NFR1	M5 citation precision, M6 unsupported-finding rate
FR13–FR14, NFR5	M8 reviewer time, M9 SUS
FR3, FR9, NFR3	Reproduced runs from logs
NFR2, NFR6	M10 latency and cost

5. Proposed Product and Innovation

5.1 Product Concept

TenderLens is a local web application. A panel member uploads the RFT and a response, reviews the extracted requirement register, starts the check, and receives an evidence matrix: one row per requirement, with a status, one to three quotations, their page locations and a short rationale. The member confirms or overrides each row and exports the result.

5.2 Major Features and User Workflow

1. Upload RFT → requirement register extracted and shown for correction.
2. Upload response → parsed into located paragraphs.

3. Run check → for each requirement, candidate passages retrieved and assessed.
4. Review matrix → each finding shows quotations highlighted in the document view.
5. Confirm or override → reasons recorded.
6. Export → spreadsheet with decisions and page references.

5.3 Innovation and Distinction from a Generic Chatbot

A chatbot answers questions about a document. TenderLens produces a complete, structured decision record in which every positive finding is backed by verified text, and every absence is explicit. Three design choices distinguish it: (a) requirement-by-requirement processing so no requirement is skipped; (b) deterministic quotation verification before anything is shown; (c) separation of AI findings from human decisions in the stored record. These choices respond directly to documented weaknesses of LLM output — unsupported claims and positional blind spots (Ji et al., 2023; Liu et al., 2024).

6. Proposed Methodology and System Design

6.1 System Architecture

The system has six components (Appendix A): a document parser (PyMuPDF), a requirement extractor (LLM with structured output, followed by human correction), a hybrid retriever (BM25 (Robertson & Zaragoza, 2009) plus dense embeddings with reciprocal-rank fusion), an assessment agent, a verifier, and a review interface (FastAPI back end, React front end, SQLite store).

6.2 Data Flow and Structured Output Contract

For each requirement the assessment agent receives the requirement text and the eight best candidate passages, each with an identifier. It must return JSON with `status` (addressed | partial | not_found), `quotes` (each with passage identifier and verbatim text) and `rationale`. The verifier checks that each quotation is a substring of the identified passage after whitespace normalisation; failing quotations are removed, and a finding left with no valid quotation is downgraded to *not found — needs manual check*. The agent may request one additional search with its own query when the first candidates are insufficient, following a bounded tool-use pattern (Yao et al., 2023) with a hard limit of two searches.

6.3 Model Strategy

Development uses a hosted model through an OpenAI-compatible interface at temperature 0. We will compare one frontier model and one smaller open-weight model on the development set and select on macro-F1 and cost. Prompts are versioned in the repository.

6.4 Development Method

Two-week iterations with a fixed demo at the end of each. Each iteration ends with a run of the evaluation harness on the development set; regressions block merging.

6.5 Data, Evaluation Material and Tools

Three public RFT templates (ICT services, facilities maintenance, consultancy) provide requirement lists. We will write 24 synthetic responses (8 per template) with deliberately varied compliance: fully compliant, indirect compliance in appendices, partial compliance and clear omissions. No real supplier data is used.

7. Business and Product Analysis

7.1 Value Proposition

For procurement panels that must justify every exclusion, TenderLens turns a linear reading task into a verification task: the reviewer checks proposed evidence instead of searching for it, and the record of where each requirement is addressed is produced as a by-product.

7.2 Intended Users and Customers

Primary users are evaluation panel members in public agencies and large private organisations with formal procurement. Customers are procurement branches, which purchase tools centrally.

7.3 Alternatives and Competitors

Alternative	Strength	Limitation for compliance checking
Manual reading with PDF search	Trusted, no new tools	Slow; keyword search misses paraphrase
E-procurement platforms	Manage lodgement and deadlines	Do not assess response content
Contract-analysis tools	Strong clause extraction	Built for contracts, not requirement-to-narrative matching
General chat assistants	Fast summaries	No verified locations; may assert unsupported compliance
TenderLens	Verified, located evidence per requirement	Requires requirement extraction step; text PDFs only

The closest competitor in capability is a general chat assistant with document upload. Our evaluation includes exactly this as a baseline (direct whole-document review) so that the benefit of verification is measured rather than asserted.

7.4 Benefits

Reduced search time; consistent coverage of every requirement; documented evidence for exclusions; faster moderation because disagreements are visible.

7.5 Feasibility and Adoption Pathway

A local prototype avoids integration barriers during evaluation. Adoption in an agency would require an approved model endpoint and a security assessment; the design keeps all data local except calls to that endpoint.

7.6 Cost and Operational Considerations

A 100-page response is approximately 60,000 tokens. With retrieval limiting each requirement's context to eight passages, a 40-requirement check uses roughly 200,000 input tokens. At current list prices for mid-tier hosted models this is below one Australian dollar per response.

8. Evaluation Plan

8.1 Dataset Construction and Labelling

The 24 synthetic responses are split into 8 development and 16 test responses, stratified by template. Two team members independently label each requirement in each response (status and evidence passages); disagreements are adjudicated by a third member. Inter-annotator agreement (Cohen's kappa) is reported before adjudication.

8.2 Baselines

- **B1 Manual review:** a team member checks test responses with PDF search only; time and decisions recorded.
- **B2 Direct LLM review:** the same model receives the full response and the full requirement list in one prompt and returns statuses and quotations, without retrieval or verification.

8.3 Test Scenarios

Fully compliant responses; evidence located only in appendices; paraphrased compliance; partial compliance; clear omission; and adversarial text (a sentence claiming "this response meets all mandatory requirements").

8.4 Metrics and Targets

ID	Metric	Definition	Target
M1	Requirement extraction accuracy	Requirements extracted with correct text and number	≥95%
M2	Locator completeness	Paragraphs with page and section	100%
M3	Evidence recall@8	Annotated evidence passages among top-8 candidates	≥0.85
M4	Status macro-F1	Agreement of status with adjudicated labels	≥0.80
M5	Citation precision	Displayed quotations judged to support the finding	≥0.95
M6	Unsupported-finding rate	Addressed findings with no valid supporting quotation	≤5% (B2 expected higher)
M7	Adversarial robustness	Adversarial sentence never cited as evidence	100%
M8	Reviewer time	Minutes per response, TenderLens vs B1	≥30% reduction
M9	Usability	SUS score	≥70
M10	Latency and cost	Per response	≤4 min; ≤A$1.50

8.5 Procedure and Reporting

Prompts are frozen before the test set is used. Each system is run three times on the test set to report variance. Reviewer-time measurement uses a within-subject design with three volunteer reviewers, counterbalancing the order of tools. Results are reported per metric with confidence intervals and with failure cases illustrated.

9. Risks, Ethics and Responsible AI

TenderLens is decision support: it never excludes a supplier, and AI findings are stored separately from confirmed decisions.

ID	Risk	L / I	Mitigation	Trigger and fallback
R1	Missed evidence leads to wrongful "not found"	M / H	Hybrid retrieval; second search allowed; "not found" always marked for manual check	Recall@8 <0.80 on dev → increase candidates to 12
R2	Fabricated or misattributed quotation	M / H	Verifier rejects non-matching quotations	Any unverified quote in review view → block release
R3	Automation bias: reviewer accepts wrong finding	M / H	Evidence shown before status; override one click	Override rate <2% in study → add confirmation step

ID	Risk	L / I	Mitigation	Trigger and fallback
R4	Confidential data leakage	L / H	Synthetic data only; provider retention disabled	Any real tender data → stop and remove
R5	Prompt injection in response text	M / M	Response treated as quoted data; adversarial tests (M7)	M7 <100% → strengthen delimiting and add filter
R6	Scope creep into scoring	M / M	Scoring explicitly out of scope	New scoring request → backlog only

10. Semester Plan and Team Responsibilities

10.1 Milestones

Week	Milestone	Deliverable
6	Parsing and requirement extraction	FR1–FR5 working on three RFTs
7	Synthetic dataset and labels	24 responses, adjudicated labels
8	Retrieval and assessment agent	FR7–FR12 on dev set; first M3–M6 results
9	Review interface	FR13–FR14; internal usability check
10	Export and comparison	FR6, FR15
11	Test-set evaluation	M1–M10 results
12	Reviewer study and refinement	M8–M9; final report draft
13	Presentation	Live demo and report

10.2 Responsibility Areas

Area	Member	Responsibilities
A1	Mia Chen	Requirement extraction, prompt versioning
A2	Liam O'Brien	Document parsing, export, dataset construction
A3	Priya Raman	Retrieval, evaluation harness
A4	Tom Nguyen	Assessment agent, verifier
A5	Sara Haddad	Review interface, reviewer study

10.3 Dependencies and De-scoping Order

Retrieval (A3) and parsing (A2) must be ready before the agent (A4) can be evaluated. If behind schedule, we de-scope in this order: FR15 comparison, FR6 export formatting, the second model comparison.

11. References

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Appendix A — System Architecture

Component	Input	Output	Technology
Document parser	RFT and response PDFs	Located paragraphs	PyMuPDF
Requirement extractor	RFT paragraphs	Numbered requirement register	LLM structured output + human correction
Hybrid retriever	Requirement, response paragraphs	Top-8 candidate passages	BM25 + embeddings, rank fusion
Assessment agent	Requirement, candidates	Status, quotations, rationale	LLM, temperature 0, ≤2 searches
Verifier	Agent output, source paragraphs	Verified findings	Deterministic substring matching
Review interface	Verified findings	Confirmed matrix, export	FastAPI, React, SQLite

Appendix B — Reviewer Workflow

The reviewer opens a response and sees the requirement list on the left and the document on the right. Selecting a requirement highlights its quotations in the document. The reviewer confirms the status or chooses another, adds a note when overriding, and moves to the next requirement. Requirements marked "not found" are listed first, because they carry the highest consequence and require the reviewer's own search before any exclusion is recorded.